

DENGQEN, China

WMO: 561160

Lat: 31.4147N

Lon: 95.5936E

Elev: 12710

StdP: 9.09

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	8.5	-17.4	3.4	26.1	-13.6	4.3	24.6	21.5	30.8	18.4	31.0	1.0	290	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	71.5	50.5	69.2	49.8	66.9	49.2	53.2	65.1	52.2	63.8	51.3	62.3	4.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.4	85.1	56.2	48.4	82.0	55.4	47.6	79.3	54.9	27.9	65.3	27.2	64.2	26.5	62.7	57.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	14.3	11.6	DB	-1.0	76.4	3.0	1.6	-3.1	77.5	-4.9	78.5	-6.5	79.3	-8.7	80.5
			WB	-2.9	55.2	2.6	1.2	-4.8	56.0	-6.3	56.8	-7.8	57.5	-9.7	58.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	23.2	27.2	32.8	40.0	47.3	53.4	56.2	55.6	51.3	41.3	31.8	25.1	(6)
(7)		DBStd	12.58	5.16	4.78	4.84	4.15	4.58	4.37	3.61	3.67	3.85	5.46	4.61	4.73	(7)
(8)		HDD50	4062	831	639	535	301	110	18	3	4	31	272	547	771	(8)
(9)		HDD65	8946	1296	1059	1000	751	549	349	274	290	412	733	997	1236	(9)
(10)		CDD50	592	0	0	0	0	26	119	194	180	69	4	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	10	0	0	0	0	0	1	4	5	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.2	3.5	3.8	3.9	3.8	3.2	2.8	2.7	2.7	2.6	2.6	(14)	
(15)	Precipitation	PrecAvg	25.0	0.1	0.3	0.5	0.9	2.1	5.0	5.5	4.8	3.9	1.5	0.3	0.1	(15)
(16)		PrecMax	34.8	0.4	0.9	1.5	2.7	4.4	8.5	8.2	10.1	6.4	3.4	1.1	0.6	(16)
(17)		PrecMin	14.9	0.0	0.0	0.0	0.1	0.3	2.0	1.9	0.9	1.5	0.3	0.0	0.0	(17)
(18)		PrecStd	4.8	0.1	0.2	0.4	0.6	1.0	1.7	1.7	2.5	1.3	0.9	0.3	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	49.3	56.8	61.3	68.4	72.9	74.1	74.4	70.3	64.5	53.1	48.7	(19)
(20)			MCWB	29.7	29.6	34.4	40.0	46.3	50.0	51.8	50.2	48.8	44.5	33.4	30.0	(20)
(21)		2%	DB	43.1	45.3	50.3	57.0	64.6	69.4	71.3	71.4	66.5	59.4	48.8	43.6	(21)
(22)			MCWB	27.4	28.7	32.8	37.6	44.0	49.2	51.3	50.2	48.1	42.4	32.1	28.0	(22)
(23)		5%	DB	38.5	41.9	46.9	53.7	61.5	66.6	68.7	68.9	63.9	55.7	45.8	39.9	(23)
(24)			MCWB	25.1	27.3	31.3	36.0	42.4	48.4	50.7	50.0	47.6	40.8	30.9	26.6	(24)
(25)		10%	DB	34.8	38.7	43.8	50.7	58.2	63.6	66.0	65.9	61.1	52.2	42.8	36.6	(25)
(26)			MCWB	23.3	25.9	30.0	35.0	41.2	48.0	50.4	49.5	47.0	39.3	29.8	24.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.1	31.8	36.7	42.3	48.7	53.2	54.8	54.2	51.8	46.4	35.7	31.0	(27)
(28)			MCDWB	46.3	45.6	51.4	58.0	63.8	65.4	67.7	66.4	64.1	59.1	46.9	45.4	(28)
(29)		2%	WB	27.8	29.6	34.3	39.5	46.3	51.9	53.6	52.9	50.3	44.4	33.6	28.7	(29)
(30)			MCDWB	41.3	42.3	47.7	53.0	60.1	64.4	65.8	64.7	61.9	56.7	45.3	42.0	(30)
(31)		5%	WB	25.6	28.0	32.4	37.7	44.6	50.7	52.6	51.9	49.2	42.5	32.0	27.0	(31)
(32)			MCDWB	37.4	39.9	43.9	49.8	57.4	62.5	64.2	63.4	60.1	52.5	43.2	38.5	(32)
(33)		10%	WB	23.7	26.5	30.9	36.3	43.0	49.5	51.7	51.0	48.2	40.8	30.6	25.3	(33)
(34)			MCDWB	33.7	37.2	41.2	47.5	54.6	60.2	62.5	61.7	58.2	49.1	40.9	35.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.5	21.6	20.0	20.1	19.7	18.5	18.2	18.9	19.4	19.3	22.2	24.6	(35)
(36)			MCDBR	30.8	28.0	25.5	25.5	25.0	23.9	23.9	25.3	24.3	24.7	27.0	29.7	(36)
(37)			MCWBR	18.5	15.7	12.2	10.7	9.0	8.0	7.6	7.9	8.8	9.6	13.1	17.2	(37)
(38)		5% WB	MCDWR	28.4	25.1	22.2	21.4	21.2	19.0	18.6	18.5	19.3	21.3	24.0	27.8	(38)
(39)			MCWBR	17.4	14.6	11.1	9.6	8.4	7.6	6.8	6.8	7.9	8.8	12.3	16.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.215	0.245	0.301	0.322	0.308	0.289	0.287	0.285	0.266	0.244	0.213	0.198	(40)	
(41)		taud	2.558	2.460	2.286	2.281	2.422	2.589	2.645	2.630	2.661	2.641	2.634	2.656	(41)	
(42)		Ebn at Noon	332	327	312	307	310	314	314	315	321	324	331	336	(42)	
(43)		Edh at Noon	25	31	40	43	37	31	30	30	27	26	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1153	1403	1689	1873	1996	1831	1796	1694	1544	1329	1266	1113	(44)	
(45)		RadStd	86	59	77	103	110	111	110	151	101	106	63	73	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.56	N/A	-3.56	N/A	-0.52	N/A	-212	-295	N/A	N/A	(46)
(47) Regional (3 neighbors)	+0.76	+1.10	-3.22	N/A	-0.41	N/A	N/A	-281	+144	N/A	(47)

Nomenclature: See separate page